MTH2008 Real Analysis: Exercise Sheet 2

This Exercise Sheet covers the material presented in Weeks 2–5. By completing the exercises, you should develop an understanding of functions of a single variable, limits and continuity, uniform continuity, and differentiability of functions of a single variable.

1. For each of the following functions f, determine when $\lim_{x\to\infty} f(x)$ exists, justifying your answer using the definition of limits at infinity.

(i) $f : \mathbb{R} \to \mathbb{R}$ defined, for $x \in \mathbb{R}$, by $f(x) = 1/(x^2+1)$

(ii) For α a non-positive real number, $f_\alpha : \mathbb{R}-\{0\} \to \mathbb{R}$ defined, for $x \in \mathbb{R}-\{0\}$, by $f_\alpha(x) = |x|^{-\alpha}\sin(x)$

2. Determine if the following functions are continuous, continuous from the right or continuous from the left at zero. Justify your answers either using the definition of a limit, or the values of known limits and the algebra of limits.

(i) $f : \mathbb{R} - \{0\} \to \mathbb{R}$ $\qquad f(x) = \dfrac{1}{x}$

(ii) $f : \mathbb{R} \to \mathbb{R}$ $\qquad f(x) = x^2$

(iii) $f : \mathbb{R} \to \mathbb{R}$ $\qquad f(x) = \begin{cases} \dfrac{x+|x|(1+x)}{x}\sin\left(\dfrac{1}{x}\right) & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$

3. Using either the definition of continuity, or known results on addition, products and compositions of continuous functions, determine all $x \in \mathbb{R}$ at which the function $x \mapsto \ln(1+\sin(x))$ is continuous.

4. Prove that if a function $f : \mathbb{R} \to \mathbb{R}$ is continuous at x_0 and $f(x_0) > \mu$ for some $x_0, \mu \in \mathbb{R}$, then $f(x) > \mu$ for all x in some open neighborhood of x_0.

5. For a set $T \subseteq \mathbb{R}$, the *characteristic function* $\psi_T : \mathbb{R} \to \mathbb{R}$ on T is defined, for $x \in \mathbb{R}$, by

$$\psi_T(x) = \begin{cases} 1 & \text{if } x \in T, \\ 0 & \text{if } x \notin T. \end{cases}$$

Prove that ψ_T is continuous at a point $x_0 \in \mathbb{R}$, if and only if $x_0 \in T^0 \cup (T^c)^0$.

6. Let f be a real valued function with domain D_f, and let $|f|$ denote the function whose value at each $x \in D_f$ is $|f(x)|$.

(i) Prove that if f is continuous at x_0, then $|f|$ is continuous at x_0.

(ii) Determine if the the converse true. Justify your answer.

7. Determine if the following functions exist.

(i) A continuous function $f\colon (0,\infty) \to \mathbb{R}$ such that f attains its minimum and its maximum, namely there exist $x_1, x_2 \in (0,\infty)$ such that $f(x_1) = \min_{x\in(0,\infty)} f(x)$ and $f(x_2) = \max_{x\in(0,\infty)} f(x)$.

(ii) A continuous and bounded function $f\colon [0,\infty) \to \mathbb{R}$ that neither attains it minimum nor its maximum.

8. Prove that if f is non-constant and continuous on a non-empty, bounded interval $I \subseteq R$, then the set $S = f(I)$ is an interval. Moreover, show that if I is a closed interval, then S is closed.

9. Let $f_1, f_2 : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$. by

$$f_1(x) = \begin{cases} x\cdot\sin(x)\cdot\sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases} \qquad \text{and} \qquad f_2(x) = \begin{cases} \dfrac{|x|\sqrt{x^2-2x+1}}{x-1} & \text{if } x \neq 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Determine, using any method, at which points $x_0 \in \mathbb{R}$ the functions f_1 and f_2 are differentiable, and for each of these points compute the derivative.

10. Let $a, b \in \mathbb{R}$ with $a < b$ be fixed and suppose that f is differentiable on the interval $[a, b]$, but f' is discontinuous at a point $x_0 \in (a, b)$. Prove that one of the following limits do not exist in $\mathbb{R}$:

$$\lim_{x \to x_0 -} f'(x) \quad \text{and} \quad \lim_{x \to x_0 +} f'(x).$$

11. Let $a, b \in \mathbb{R}$ be fixed and let $f, g : \mathbb{R} \to \mathbb{R}$ be defined, for $x \in \mathbb{R}$, by

$$f(x) = x^3 + ax + b \quad \text{and} \quad g(x) = x^4 + ax + b.$$

(i) Prove that there exists $c \in \mathbb{R}$ with $f(c) = 0$.

(ii) Prove that there exist at most two real numbers $\alpha_1, \alpha_2 \in \mathbb{R}$ such that $g(\alpha_1) = g(\alpha_2) = 0$.

12. Suppose that $f, g : \mathbb{R} \to \mathbb{R}$ are differentiable at some $x_0 \in \mathbb{R}$ with $f(x_0) = g(x_0) = 0$ and $g'(x_0) \neq 0$. Without using L'Hôpital's rule, prove that,

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{f'(x_0)}{g'(x_0)}.$$